Space Radiation, Diagnostic X-rays, and Radiation Interaction with Biological Materials

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Q1. Types of Space Radiation

Space radiation consists primarily of Galactic Cosmic Rays (GCRs), Solar Particle Events (SPEs), Van Allen belt particles, and secondary radiation produced when primary particles interact with spacecraft materials.

Galactic Cosmic Rays (GCRs)

GCRs are composed mainly of protons ($\sim 85\%$), alpha particles ($\sim 14\%$), and heavy ions (HZE particles). Their energies range from

$$10^8 \text{ eV to } 10^{20} \text{ eV}.$$

Using Planck's equation,

$$E = h\nu,$$

the equivalent photon frequency is

$$\nu = \frac{E}{h}.$$

For a 1 GeV particle,

$$\nu \approx 2.4 \times 10^{23} \text{ Hz}.$$

Biological effects include DNA double-strand breaks, chromosomal aberrations, cancer induction, cataracts, cardiovascular disease, and neurological damage.

Solar Particle Events (SPEs)

Solar particle events are generated during solar flares and coronal mass ejections. They mainly contain protons with energies from

$$10 \text{ MeV} - 500 \text{ MeV},$$

corresponding to frequencies of approximately

$$10^{21} - 10^{23} \text{ Hz}.$$

Large SPEs may produce acute radiation syndrome during long-duration missions.

Van Allen Radiation Belts

The Earth's magnetic field traps energetic protons and electrons. The inner belt is proton-rich while the outer belt contains energetic electrons. These particles are a major concern during lunar missions crossing the belts.

Secondary Radiation

Primary cosmic rays striking spacecraft materials produce neutrons, gamma rays, muons, pions, and secondary electrons that contribute significantly to astronaut dose.

Typical Astronaut Dose

Mission	Effective Dose
Natural background	2–3 mSv/year
ISS (6 months)	80–160 mSv
Lunar mission (6 months)	200–400 mSv
Mars mission (6 months)	600–1200 mSv

Q2. Working of Diagnostic X-ray Machines

A diagnostic X-ray tube consists of a cathode, tungsten filament, focusing cup, evacuated tube, tungsten anode, high-voltage supply, protective housing, and X-ray window.

Electron Emission

Heating the tungsten filament produces thermionic emission described approximately by the Richardson–Dushman equation,

$$J = AT^2 e^{-\phi/kT}.$$

Electron Acceleration

Electrons are accelerated through a potential difference of approximately 30–150 kV.

Their kinetic energy is

$$K = eV.$$

For a 100 kV tube,

$$K = 100 \text{ keV}.$$

X-ray Production

Two mechanisms produce X-rays:

1. Bremsstrahlung radiation from electron deceleration near nuclei.
2. Characteristic X-rays produced when inner-shell vacancies are filled by outer-shell electrons.

The maximum Bremsstrahlung energy is

$$E_{\text{max}} = eV.$$

Frequency Range Used

Diagnostic X-rays typically have energies between

$$20 - 150 \text{ keV},$$

corresponding to frequencies of

$$5 \times 10^{18} - 4 \times 10^{19} \text{ Hz},$$

and wavelengths between approximately

$$0.008 - 0.06 \text{ nm}.$$

This range provides sufficient penetration through soft tissue while maintaining contrast between tissues and bone.

Radiation Damage

X-rays damage tissue through direct DNA ionization and indirect free-radical production.

Water radiolysis occurs according to

$$H_2O \to OH^\bullet + H^\bullet + e^-.$$

Deterministic effects include skin burns and cataracts, while stochastic effects include cancer and hereditary mutations.

Q3. Interaction of Radiation with Biological Material

Ionizing radiation interacts with tissue through photoelectric absorption, Compton scattering, pair production, and Coulomb interactions.

Direct Action

DNA may be directly ionized:

$$DNA \to DNA^+ + e^-.$$

This can produce strand breaks and chromosomal damage.

Indirect Action

Radiolysis of water generates highly reactive hydroxyl radicals:

$$H_2O \to OH^\bullet + H^\bullet + e^-.$$

These radicals attack DNA, proteins, and cell membranes.

Cellular Effects

Biological effects include apoptosis, necrosis, mutation, carcinogenesis, and tissue dysfunction. Rapidly dividing tissues such as bone marrow and gastrointestinal epithelium are particularly radiosensitive.

Radiation Dose Units

Exposure

$$X = \frac{dQ}{dm},$$

with SI unit C/kg.

Absorbed Dose

$$D = \frac{dE}{dm},$$

measured in Gray (Gy), where

$$1 \text{ Gy} = 1 \text{ J/kg}.$$

Equivalent Dose

$$H = D w_R,$$

measured in Sievert (Sv).

Typical radiation weighting factors are:

Radiation	w_R
X-rays	1
Gamma rays	1
Beta particles	1
Protons	2
Alpha particles	20

Effective Dose

$$E = \sum w_T H_T,$$

also measured in Sievert.

Importance of Dosimetry

Radiation dosimetry ensures patient safety, treatment accuracy, occupational protection, environmental monitoring, astronaut health, and compliance with international standards. The ALARA (As Low As Reasonably Achievable) principle is used to minimize unnecessary radiation exposure.